

■ AmritChidiya — Comprehensive 4-Member Team Role Division & Ownership Handbook

Project: AmritChidiya (■■■■■■■■■■) — "Apni Sone Ki Chidiya Ko Phir Se Udaan Do" ■■

Document Purpose: Complete technical specification and task division breaking down codebase ownership, step-by-step execution for each role, viva talking points, and standout technical USPs across **4 team members** (explained so clearly that any beginner can understand).

1. The Big Picture: How the 4 Members Work Together (For Beginners)

Think of AmritChidiya like a modern, highly secure digital government office. Each member is the sole engineer and owner of one critical department:

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M1["Member 1: The Citizen Counter<br><b>Frontend UI/UX & Citizen Access Lead</b><br>USP: Phygital CSC Dossier Bridge"]
M2["Member 2: The Ears & Speaking Voice<br><b>Backend API & Speech Processing Lead</b><br>USP: Low-Latency Vernacular Audio Pipeline"]
M3["Member 3: The Scheme Expert Brain<br><b>AI Agent, Scheme Logic & Cyber Safety Lead</b><br>USP: Anti-Hallucination Matcher & Cyber Shield"]
M4["Member 4: The Bank Vault & Cloud Guard<br><b>Database, Auth Security & DevOps Lead</b><br>USP: Zero-Trust Hybrid Cloud & Bcrypt Auth"]

M1 -->|HTTP POST /chat (JWT Protected)| M2
M1 -->|Audio Upload /transcribe & /tts| M2
M2 -->|Invoke State Machine| M3
M3 -->|Resolve Scheme URLs & Cyber Rules| M2
M2 -->|Supabase / SQLite Failover CRUD| M4
M1 -->|Client JWT & Session Sync| M4
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- **Member 1 (The Front Counter):** Builds the welcoming touch-screen interface where the citizen talks, checks their eligibility, and receives a printed receipt to take home.
- **Member 2 (The Ears & Voice):** Listens to the citizen's voice in regional Indian accents, converts it into text at lightning speed, and speaks back with a human-sounding voice.
- **Member 3 (The Scheme Expert & Security Detective):** Knows every single government rule, guarantees 0% false promises, and checks suspicious WhatsApp links to protect citizens from scams.
- **Member 4 (The Bank Vault & Cloud Guard):** Locks up user passwords with unhackable 12-round Bcrypt encryption, gives out secure digital entry passes (JWT tokens), and keeps a backup database running 24/7.

2. Technical Responsibility & Deliverables Matrix

Member	Department & Role	Core Assigned Files	Member Standout Technical USP
Member 1	Frontend UI/UX & Citizen Access Lead	frontend/app/page.tsx frontend/components/CscDossierModal.tsx frontend/components/EligibilityCalculatorModal.tsx frontend/components/MarkdownContent.tsx frontend/app/globals.css	Phyigital Last-Mile Bridge & Citizen CSC Dossier Generator Bridges the digital-to-physical divide for rural India by generating structured, printable CSC Scheme Dossiers (AC-CSC-XXXXXX) with NPCI/DBT bank checklists & VLE operator sign-off boxes for village Jan Seva Kendras, coupled with a responsive Next.js 16 UI and hands-free Talk Mode.

Answer: Groq runs on specialized LPU (Language Processing Unit) chips, providing sub-500ms transcription latency. This makes voice conversations feel real-time rather than having an awkward 3-second delay.

- **Q2: How do you protect user privacy for voice recordings?**

Answer: Audio files are treated as ephemeral streams. They are created in temporary OS memory/disk via `tempfile.NamedTemporaryFile` and unconditionally unlinked (`os.unlink()`) inside a `finally` block immediately after transcription finishes.

- Q3: How does the text sanitization function work before TTS?

Answer: `clean_text_for_speech()` uses regular expressions to strip out markdown asterisks, hashes, code blocks, and URLs. It turns bullet points into natural sentence pauses so the voice sounds like a caring human companion.

5. Member 3: AI Agent, Scheme Logic & Cyber Safety Lead

■ Member USP (Unique Selling Proposition)

"Anti-Hallucination Welfare Match Engine, Domain Cyber Shield & National Helpline Bridge"

Why this matters for a beginner: If an AI makes up a fake scholarship or sends a student to an expired link, citizens lose faith in the system. Member 3 built a strict LangGraph state machine that prohibits fake schemes, combined with the **Official Domain Shield & Scam Checker** (ScamShieldModal.tsx) that verifies authentic .gov.in domains, teaches the 4 Golden Rules of fraud defense, links directly to the National Cyber Crime Helpline (1930), and protects citizens from fake WhatsApp schemes demanding registration fees.

Assigned Codebase Files

- [backend/agents/chat_agent.py](#) — LangGraph state graph, scheme extraction regex engine, official government URL resolver.
- [backend/prompts/system_prompt.py](#) — Core system prompt, role confinement, phase transition protocols, hyperlinked scheme markdown templates.
- [frontend/components/ScamShieldModal.tsx](#) — Official domain verification tool, 4-rule scam detector, and 1930 Cyber Helpline bridge.

■ Step-by-Step Working Mechanism

- 1. Phase 1 (Profile Intake):** If a student says *"Mujhe scholarship chahiye"*, the agent checks if State, Income, Category, and Education are provided. If missing, it asks polite clarifying questions and suppresses scheme extraction.
- 2. Phase 2 (Eligibility Delivery):** Once details are verified, the agent outputs exact scheme names formatted in structured markdown headers.
- 3. Government URL Mapper & Fallback:** `SCHEME_URL_MAP` in `chat_agent.py` maps each scheme name directly to official government portals (`scholarships.gov.in`, `pmkisan.gov.in`, `scholarship.up.gov.in`). If an unmapped scheme is detected, it securely defaults to `https://www.myscheme.gov.in/search?q=...` (India's official National Scheme Portal), guaranteeing zero phishing links.
- 4. Phase 3 (Step-by-Step Roadmap):** Delivers a 5-step registration breakdown with required document checklists.
- 5. Cyber Shield & Fraud Education:** When a citizen pastes a WhatsApp link into `ScamShieldModal.tsx`, the code extracts the hostname, checks for official `.gov.in/.nic.in` accreditation, flags phishing TLDs (`.xyz`, `.top`), and displays the 4 Golden Rules with a direct dialer to 1930.

■ Interview / Viva Questions & Model Answers

- **Q1: How does LangGraph prevent LLM hallucinations in government schemes?**

Answer: LangGraph enforces a structured state machine. The prompt explicitly prohibits suggesting schemes during profile intake, and the regex extractor only accepts schemes formatted in strict headers that match verified criteria.

- **Q2: How does the Cyber Shield protect against WhatsApp fraud?**

Answer: It inspects the domain structure. All genuine Indian central and state welfare portals must end in **.gov.in** or **.nic.in**. If a link ends in **.xyz** or asks for private UPI fees, it triggers a high-risk scam alert and points the citizen to the National Cyber Crime Helpline (1930).

